

Course Title	ENGINEERING CHEMISTRY			Course Type		Theory and Integrated Lab			
Course Code	24CH110/ 24CH210	Credits	L : T : P 3 : 0 : 2	Class		I/II Semester			
Course Structure	TLP	Credits	Contact Hours /week	Total Number of Classes / Semester		Assessment in Weightage and Marks			
	Theory	3	3	L	P	CIE		SEE	
	Practical	1	2	--	--	Weightage	40%	Weightage	60%
						Max. Marks	40 Marks	Max. Marks	100 Marks
	Total	4	5	40	26	Min. Marks	20 Marks	Min. Marks	41 Marks

Note: For passing the student has to score a minimum of 45 marks (CIE+SEE: 20+25 or 21+24)

COURSE PREREQUISITES: NIL

COURSE OBJECTIVE:

The objectives of this course are -

1. To impart the knowledge of applied aspects of chemistry.
2. To utilize the same for the technological advancement in various discipline of engineering.
3. To impart practical skills in the field of chemical analysis.

COURSE OUTCOMES (COs)

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Apply the concepts of electrochemistry to explain the electrochemical phenomenon involved in energy storage and conversion devices.	3
CO2	Explain the mechanism of corrosion and its prevention.	2
CO3	Explain the chemistry and applications of polymeric materials.	3
CO4	Explain the importance of petroleum products as conventional sources of energy.	3
CO5	Determine the composition and quality parameters of samples by conventional and instrumental techniques.	3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

UNIT 3 V	POLYMER MATERIAL Introduction, Thermoplastics and thermosetting plastics with examples, weight average and number average molecular weight, Numerical problems. Glass transition temperature (T_g) – parameters affecting T_g and significance of T_g. Synthesis, properties and applications of PTFE, PMMA and PU. Elastomers: definition, deficiencies of natural rubber, advantages of synthetic rubber. Synthesis and applications of neoprene and butyl rubber. Biodegradable polymers: Introduction and their requirements. Synthesis and properties of polylactic acid, Applications of biodegradable polymers in medical industry. Photoconducting polymers: definition, poly vinyl carbazole- synthesis and its applications.	8
UNIT 4 VI	ENERGY SOURCES Chemical fuels: Definition, classification; calorific value-definition, Gross and Net calorific values (SI units). Determination of calorific value of a solid/liquid fuel using Bomb calorimeter, numerical problems, Petroleum cracking- fluidized catalytic cracking. Reformation of petrol. Knocking and its mechanism, octane number, cetane number, prevention of knocking, anti-knocking agents, unleaded petrol.	6
UNIT 5 I III	CONVENTIONAL AND ANALYTICAL TECHNIQUES FOR CHEMICAL ANALYSIS Conventional methods: Determination of total hardness of water by Complexometric Method. Determination of total alkalinity of water sample and identification of the type and extent of alkalinity. Determination of dissolved oxygen of the given water sample by Winkler's method. Determination of Chemical Oxygen Demand of an industrial effluent. Determination of amount of Calcium oxide in cement solution. Numerical Problems. Instrumental techniques: Estimation of FAS Potentiometrically using standard solution of potassium dichromate. Determination of pK_a of weak acid using pH meter. Conductometric estimation of HCl and CH₃COOH present in an acid mixture using standard solution of sodium hydroxide. Determination of copper colorimetrically using ammonia as the complexing agent. Determination of iron (III) by colorimetric method using potassium thiocyanate as the complexing agent.	10

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	3	-	-	-
CO2	3	3	-	-	-	-	2	-	-	-	-	-
CO3	3	3	-	-	-	-	-	-	3	-	-	-
CO4	3	3	-	-	-	-	2	-	3	-	-	-
CO5	3	3	-	-	-	-	-	-	3	-	-	-

Course Content / Syllabus:

UNIT No.	Content	Hours
UNIT 1 <u>II</u>	ELECTROCHEMISTRY Electrode system: Introduction, Single electrode potential- formation of a cell, representation of cell, emf of a cell, Nernst equation - Derivation and numerical problems. Secondary reference electrodes - definition, Calomel electrode-Construction, working and applications, Ion selective electrode- definition, construction working and determination of pH using glass electrode. Battery technology: Introduction, definition, battery characteristics, classification primary, secondary and reserve batteries with examples. Modern batteries-construction, working and applications of Nickel-Metal hydride, Lithium-MnO₂ and Li-ion batteries.	8
UNIT 2 <u>IV</u>	CORROSION AND ITS PREVENTION Corrosion- definition, types-chemical and electrochemical corrosion. Electrochemical theory of corrosion, Factors affecting the rate of corrosion- nature of metal, nature of corrosion product, relative areas of anode and cathode, temperature, pH. Types of corrosion- differential metal corrosion, differential aeration corrosion (pitting and waterline corrosion). Corrosion control- Metal coating- galvanizing and tinning, Cathodic protection- sacrificial anode and impressed current techniques. Corrosion inhibitors- anodic and cathodic inhibitors with example.	8